

Guide to Typst

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tschinz

whynotlogic@gmail.com

HEI-Vs

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1 | Introduction

The goal of this document is to have the most common used elements for the markup language typst readily available. A detailed documentation can be found on their website: <https://typst.app/docs> It is to note that these are **my** most common used elements. For some elements custom templates are needed:

- [tablex](#)
- [myref](#)
- all files in the 00-templates/ folder such as
 - boxes.typ
 - constants.typ
 - helpers.typ
 - items.typ
 - metadata.typ
 - template-*

2 | Installation

2.1 With cargo

If you use already the rust programming language then you can use rust to install the latest toolchain.

```
# install rust and cargo
curl https://sh.rustup.rs -sSf | sh

# install typst
cargo install --git https://github.com/typst/typst
```

2.2 MacOS

On MacOS you can sue homebrew

```
# install homebrew
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# install typst
brew install typst
```

2.3 Linux

In Linux you can use the commonly available package manager

```
brew install typst
pacman -S typst
xbps-install typst
sudo apt-get install typst
```

3 | Formatting

3.1 Markup

Name	Example	Raw
Singleline Comment		//
Multiline Comment		/* */
Paragraph break		blankline
Line break		\
bold	bold	*bold*
italic	<i>italic</i>	_italic_
monospaced	monospaced	`monospaced`
math	$x = 1$	$x=1$
lowercase	lower	#lower("LoWeR")
uppercase	UPPER	#upper("UpPeR")
smallcaps	SMALLCAPS	#SmallCaps("smallcaps")
smartquote	“test”	#smartquote()test#smartquote()
overline	<u>overline</u>	#overline("overline")
underline	<u>underline</u>	#underline("underline")
strike	strike	#strike("strike")
sub	Text _{sub}	Text#sub("sub")
super	Text ^{super}	Text#super("super")
Label		<label>
Reference		@label

3.2 Page Formatting

```
#pagebreak() // pagebreak
#parbreak()  // parbreak
\           // linebreak
```

3.3 Space

A

B

A #h(5cm) B,

C

D

C #v(0.2cm) D

3.4 Text Formatting

For the custom textsizes and colors you need to import:

```
#import "../01-tail/constants.typ": *
```

Name	Example	Raw
Sizes	8pt text tiny text	text(8pt, "8pt text") text(tiny "tiny text")
	9pt text smaller text	text(9pt, "9pt text") text(smaller "smaller text")
	10pt text small text	text(10pt, "10pt text") text(small "small text")
	11pt text normal text	text(11pt, "11pt text") text(normal "normal text")
	14pt text large text	text(14pt, "14pt text") text(large "large text")
	16pt text larger text	text(16pt, "16pt text") text(larger "larger text")
	24pt text huge text	text(24pt, "24pt text") text(huge "huge text")
Types	36pt text huger text	text(36pt, "36pt text") text(huger "huger text")
	Fira Sans	text(font:"Fira Sans", "Fira Sans")
	Fira Mono	text(font:"Fira Mono", "Fira Mono")
	Source Sans Pro	text(font:"Source Sans Pro", "Source Sans Pro")
	New Computer Modern	text(font:"New Computer Modern", "New Computer Modern")
	New Computer Modern Sans	text(font:"New Computer Modern Sans", "New Computer Modern Sans")

Alignment	start	<code>align(start){start}</code>
	end	<code>align(end){end}</code>
	left	<code>align(left){left}</code>
	center	<code>align(center){center}</code>
	right	<code>align(right){right}</code>
	top	<code>align(top){top}</code>
	horizon	<code>align(horizon){horizon}</code>
	bottom	<code>align(bottom){bottom}</code>
	center + horizon	<code>align(center + horizon){center + horizon}</code>

Colors	black	#text(fill:black)[black]
	red	#text(fill:red)[red]
	green	#text(fill:green)[green]
	blue	#text(fill:blue)[blue]
	purple	#text(fill:purple)[purple]
	gray-80	#text(fill:gray-80)[gray-80]
	gray-70	#text(fill:gray-70)[gray-70]
	gray-60	#text(fill:gray-60)[gray-60]
	gray-50	#text(fill:gray-50)[gray-50]
	gray-40	#text(fill:gray-40)[gray-40]
	gray-30	#text(fill:gray-30)[gray-30]
	gray-20	#text(fill:gray-20)[gray-20]
	gray-10	#text(fill:gray-10)[gray-10]
	hei-orange	#text(fill:hei-orange)[hei-orange]
	hei-blue	#text(fill:hei-blue)[hei-blue]
	hei-pink	#text(fill:hei-pink)[hei-pink]
	hei-yellow	#text(fill:hei-yellow)[hei-yellow]
	hei-green	#text(fill:hei-green)[hei-green]
	spl-green	#text(fill:spl-green)[spl-green]
	spl-blue	#text(fill:spl-blue)[spl-blue]
	spl-pink	#text(fill:spl-pink)[spl-green]
	color-info	#text(fill:color-info)[color-info]
	color-idea	#text(fill:color-idea)[color-idea]
	color-warning	#text(fill:color-warning)[color-warning]
	color-important	#text(fill:color-important)[color-important]
	color-fire	#text(fill:color-fire)[color-fire]
	color-rocket	#text(fill:color-rocket)[color-rocket]
	color-todo	#text(fill:color-todo)[color-todo]
	code-bg	#text(fill:code-bg)[code-bg]
	code-border	#text(fill:code-border)[code-border]

4 | Elements

4.1 Headings

```
= Heading 1
== Heading 1.1
=== Heading 1.1.1
==== Heading 1.1.1.1
...
```

4.2 Lists

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
 - Second
 - Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
list(
  [First],
  [Second],
  [Third],
)
```

1. First
 1. Second
2. Third

```
+ First
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

Text

4. Fourth
5. Fifth

1. First
 - a) Second
2. Third

```
+ First
#set enum(numbering: "a")
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

Text

4. Fourth
5. Fifth

4.3 Custom Lists

```
#import "../01-head/items.typ": *
```

- ☑ item-list
- ⊛ item-checkbadge
- ⊙ item-checkcircle
- ☑ item-checksquare
- ✓ item-check
- 📁 item-file
- 📁 item-folder
- ⊗ item-xcircle
- ⊗ item-xsquare
- × item-x

```
#item-list(content:"item-list")
#item-checkbadge(content:"item-checkbadge")
#item-checkcircle(content:"item-checkcircle")
#item-checksquare(content:"item-checksquare")
#item-check(content:"item-check")
#item-file(content:"item-file")
#item-folder(content:"item-folder")
#item-xcircle(content:"item-xcircle")
#item-xsquare(content:"item-xsquare")
#item-x(content:"item-x")
```

4.4 Images

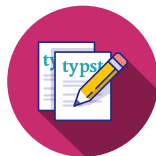
4.4.1 Alignment

left



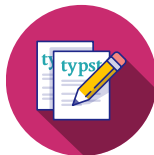
```
#image("../04-resources/icon.svg",
width: 2cm)
```

center



```
#align(center,
image("../04-resources/icon.svg",
width: 2cm)
)
```

right



```
#align(right,
image("../04-resources/icon.svg",
width: 2cm)
)
```

4.4.2 Caption

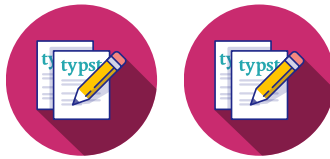


Figure 1: ZNotes Icon

```
#figure(
  image("../04-resources/icon.svg",
    width: 2cm),
  caption: [ZNotes Icon]
) <fig-icon>
```

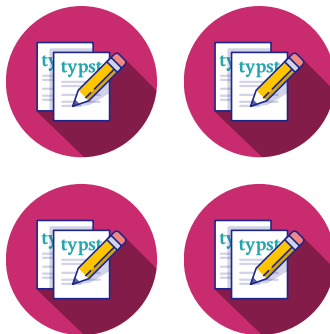
4.4.3 Cluster

Two images one caption

Figure 2: Multiple images **one** caption

```
#figure(
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm)
  ),
  caption: [Multiple images *one* caption]
)
```

Four images one caption

Figure 3: Multiple images **one** caption

```
#figure(
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm),
    image(icon, width: 2cm), image(icon, width: 2cm),
  ),
)
```

```
caption: [Multiple images *one* caption]
)
```

Two images two caption

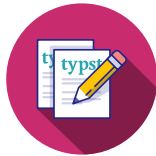


Figure 4: Caption left image



Figure 5: Caption right image

```
#align(center,
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption left image]), figure(image(icon,
width: 2cm), caption: [Caption right image]),
  ))
```

Four images four caption

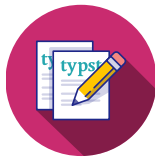


Figure 6: Caption topleft image

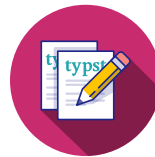


Figure 7: Caption topright image

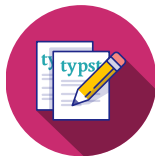


Figure 8: Caption bottomleft image



Figure 9: Caption bottomright image

```
#align(center,
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption topleft image]),
    figure(image(icon, width: 2cm), caption: [Caption topright image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomleft image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomright image]),
  ))
```

4.5 Tables

For all `#tablex` command the appropriate module needs to be imported

```
#import "../01-head/tablex.typ": *
```

Tables with and without caption

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

Table 1: Table caption

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
  ]  
)
```

```
figure(  
  tablex(  
    columns: 3,  
    align: center + horizon,  
    [  
      , [*Col1*] , [*Col2*],  
    [*Row1*], "cell-0-0", "cell-1-0",  
    [*Row2*], "cell-0-1", "cell-1-1",  
    ],  
    kind: table,  
    caption: [Table Caption]  
  )  
)
```

Tables with cell spans

	Col1	Col2
Row1	cell-0	cell-1-0
Row2		cell-1-1

	Col1	Col2
Row1	cell-0	
Row2	cell-0-1	cell-1-1

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], rowspanx(2)[cell-0],  
  "cell-1-0",  
  [*Row2*], "cell-1-1",  
  ]  
)
```

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], colspanx(2)[cell-0],  
  [*Row2*], "cell-0-1", "cell-1-1",  
  ]  
)
```

Table Design

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
tablex(  
  columns: 3,
```

```
tablex(  
  columns: 3,
```

```
auto-vlines: false,
align: center + horizon,
[
  , [*Col1*] , [*Col2*],
[*Row1*], "cell-0-0", "cell-1-0",
[*Row2*], "cell-0-1", "cell-1-1",
)
```

```
auto-hlines: false,
align: center + horizon,
[
  , [*Col1*] , [*Col2*],
[*Row1*], "cell-0-0", "cell-1-0",
[*Row2*], "cell-0-1", "cell-1-1",
)
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
#tablex(
  columns: 3,
  auto-lines: false,
  align: center + horizon,
  (), vlinex(stroke: blue) , vlinex(), (),
  [
    , [*Col1*] , [*Col2*], hlinex(stroke: red),
  [*Row1*], "cell-0-0", "cell-1-0", hlinex(),
  [*Row2*], "cell-0-1", "cell-1-1",
  )
```

4.6 Icon Boxes

#import "../01-head/boxes.typ": *



#infobox()["infobox"]



#ideabox()["ideabox"]



#warningbox()["warningbox"]



#importantbox()["importantbox"]



```
#firebox()["firebox"]
```



```
#rocketbox()["rocketbox"]
```



```
#todobox()["todobox"]
```



```
#iconbox(icon:"../04-resources/placeholder.svg", linecolor:
hei-blue)["iconbox"]
```

```
#iconbox(linecolor: hei-pink)["iconbox without icon"]
```

4.7 Color Boxes

```
#import "../01-head/boxes.typ": *
```

Exercise

Some text

```
#colorbox( title: "Exercise", color:hei-blue)[Some text]
```

Attention

Some text

```
#colorbox( title: "Attention", color:hei-pink)[Some text]
```

Consider

Some text

```
#slantedColorbox( title: "Consider", color:hei-green)[Some text]
```

Information

Some text

```
#slantedColorbox( title: "Information", color:hei-orange)[Some text]
```

4.8 Title Box

```
#import "../01-head/sections.typ": *
```

Title

Subtitle

```
#titlebox(title:[Title], subtitle:[Subtitle])
```

Title

Subtitle

```
#titlebox(width:50%, radius:0pt, border:1pt, linecolor: hei-blue, titlesize: larger,
subtitle: large, title:[Title], subtitle:[Subtitle])
```

Title

```
#titlebox(linecolor: hei-green, titlesize: larger, subtitle: large, title:[Title])
```

4.9 Exam Header

```
#import "../01-head/sections.typ": *
```

Name: -----

```
#exam_header(nbrEx:0, lang: "en")
```

Name: -----

Grade

```
#exam_header(nbrEx:1, lang: "en")
```

Name: -----

1	Grade
(10)	

```
#exam_header(nbrEx:2, pts:10, lang: "en")
```

Name: -----

1	2	Grade
(10)	(10)	

```
#exam_header(nbrEx:3, pts:10, lang: "en")
```

Name: -----

1	2	3	Grade
(10)	(10)	(10)	

```
#exam_header(nbrEx:4, pts:10, lang: "en")
```

Name: -----

1	2	3	4	Grade
(10)	(10)	(10)	(10)	

```
#exam_header(nbrEx:5, pts:10, lang: "en")
```

1 2 3 4 5 Grade

Name:

(10)	(10)	(10)	(10)	(10)	

#exam_header(nbrEx:6, pts:10, lang: "en")

	1	2	3	4	5	6	Grade
(10)	(10)	(10)	(10)	(10)	(10)	(10)	

#exam_header(nbrEx:7, pts:10, lang: "en")

	1	2	3	4	5	6	7	Grade
(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	

#exam_header(nbrEx:8, pts:10, lang: "en")

	1	2	3	4	5	6	7	8	Grade
(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	

#exam_header(nbrEx:9, pts:10, lang: "en")

	1	2	3	4	5	6	7	8	9	Grade
(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	

#exam_header(nbrEx:10, pts:10, lang: "en")

4.10 Exam Reminder

#import "../01-head/sections.typ": *

Exam Reminder:

You can only use the following items:

- the two-page summary you created.
- a pocketcalculator



Kommentieren Sie ausserdem den gesamten High-Level- und Assembler-Code ordnungsgemäss aus, um seinen Zweck und seine Einbindung in die Programmstruktur zu erklären.

Viel Glück!

```
#exam_reminder_isc(lang: "en")
```

Prüfungserinnerung:

Sie können nur die folgenden Elemente verwenden:

- die zweiseitige Zusammenfassung, die Sie erstellt haben.
- einen Taschenrechner



Kommentieren Sie ausserdem den gesamten High-Level- und Assembler-Code ordnungsgemäss aus, um seinen Zweck und seine Einbindung in die Programmstruktur zu erklären.

Viel Glück!

```
#exam_reminder_isc(lang: "de")
```

Rappel d'examen :

Vous ne pouvez utiliser que les éléments suivants :

- le résumé de deux pages que vous avez créé.
- une calculatrice de poche



In addition, properly comment all high-level and assembler code to explain its purpose and how it fits into the program structure.

Good Luck!

```
#exam_reminder_isc(lang: "fr")
```

5 | References

5.1 Links

Example	Raw
https://example.com	<code>https://example.com</code>
https://example.com	<code>#link("https://example.com")</code>
See example.com	<code>#link("https://example.com")[See example.com]</code>
whynotlogic@gmail.com	<code>#link("mailto:whynotlogic@gmail.com")[whynotlogic\@gmail.com]</code>
	<code>#link("https://tschinz.github.io/znotes")[#image(icon, width:0.5cm)]</code>

Table 2: Links

5.2 Crossreferences

In the document the following references were added.

```
= References <sec-ref>
== Links <sec-links>
#figure(image("../04-resources/icon.svg", width: 2cm)) <fig-icon>
#figure(tablex(...), kind:table) <tab-links>
#figure(align(left, raw(...)) <code-ref>
$ sum_(k=1)^n k = (n(n+1)) / 2 $ <math-eq1> #ref(<math-eq1>)
```

Listing 1: Label inserts

They can be references as follows:

Type	Example	Raw
Section	Section 5	<code>@sec-ref</code>
Subsection	Section 5.1	<code>@sec-links</code>
Table	Table 2	<code>@tab-links</code>
Code	Listing 1	<code>@code-ref</code>

5.3 External References

Example

[1]

Raw

```
#cite("stateoftheArt")
```

[1]

```
@stateoftheArt
```

5.4 Glossary

The glossary entries need to be defined in `03-tail/glossary.typ`. For the glossary functions the “import” of `01-head/helpers.typ` is needed.

```
#import "../01-head/helpers.typ": *
#import "../03-tail/glossary.typ": *
```

Example**Raw**

Scrum

```
#gls-scrumm.name
```

Scrum is an agile process framework for managing complex knowledge work, with an initial emphasis on software development, although it has been used in other fields and is slowly starting to be explored for other complex work, research and advanced technologies.

```
#gls-scrumm.description
```

5.5 Acronym

The acronym entries need to be defined in 03-tail/glossary.typ. For the acronym functions the “import” of 01-head/helpers.typ is needed.

```
#import "../01-head/helpers.typ": *
#import "../03-tail/glossary.typ": *
```

Example**Raw**

AR

```
#acr-ar.abr
```

AR

```
#acrshort(acr-ar.abbr)
```

Augmented Reality

```
#acr-ar.long
```

Augmented Reality)

```
#acrlong(acr-ar)
```

Augmented Reality (AR)

```
#acr-ar.long (#acr-ar.abbr)
```

Augmented Reality (AR)

```
#acrfull(acr-ar)
```

6 | Code

inline monospaced string

```
fn main() {println!("Hello world!")}
```

```
fn main() {  
  println!("Hello world!")  
}
```

```
fn main() {  
  println!("Hello world!")  
}
```

Listing 2: Rust Code

```
`inline monospaced string`
```

```
raw(lang:"rust",  
  "fn main() {println!(\"Hello world!  
  \")\"  
  )  
)
```

```
``rust  
fn main() {  
  println!("Hello world!")  
}  
``
```

```
#figure(  
  align(left,  
    ``rust  
    fn main() {  
      println!("Hello world!")  
    }  
    ``  
  ),  
  caption: [Rust Code],  
)
```

7 | Math Equations

Inline math

Let a and b , and c be the side of a right-angled triangle.

Let a and b , and c be the side of a right-angled triangle.

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$\sum_{k=1}^n k = (n(n+1)) / 2$,

Fullline math

$$a^2 + b^2 = c^2 \tag{1}$$

$a^2 + b^2 = c^2$ `<math-eq1>`

Math with caption

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{2}$$

Figure 10: Some proof

```
#figure(
  $ \sum_{k=1}^n k = (n(n+1)) / 2 $,
  caption: [Some proof]
)
```

7.1 Align

Formula

$$\begin{aligned} a_1 &= b_1 + c_1 = z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{3}$$

Raw

```
$
a_1 = b_1 + c_1 = z_1 \ \
a_2 = b_2 + c_2 - d_2 + e_2 = z_1
$
```

$$\begin{aligned} a_1 &= b_1 + c_1 &&= z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{4}$$

```
$
a_1 &= b_1 + c_1 &= z_1 \ \
a_2 &= b_2 + c_2 - d_2 + e_2 &= z_1
$
```

7.2 Symbols

This is an incomplete list for all symbols goto [here](#)

Outside of the $\$$ math environment the symbols can be accessed with `.`

7.2.1 Accents

Symbol Raw $\grave{x}$ `$grave(x)$` $\tilde{x}$ `$tilde(x)$` $\ddot{x}$ `$dot.double(x)$` $\ddot{x}$ `$diaer(x)$` $\check{x}$ `$caron(x)$` $\cancel{x}$ `$cancel(x)$` $\overline{xyz}$ `$overline(xyz)$` $\underbrace{xyz}$ `$underbracket(xyz)$`**Symbol Raw** $\acute{x}$ `$acute(x)$` $\breve{x}$ `$breve(x)$` $\ddot{x}$ `$dot.triple(x)$` $\mathring{x}$ `$circle(x)$` $\vec{x}$ `$arrow(x)$` $\bar{x}$ `$macron(x)$` $\underbrace{xyz}$ `$underbrace(xyz)$` $\overbrace{xyz}$ `$overbracket(xyz)$`**Symbol Raw** $\hat{x}$ `$hat(x)$` $\dot{x}$ `$dot(x)$` $\ddot{x}$ `$dot.quad(x)$` $\ddot{x}$ `$acute.double(x)$` $\vec{x}$ `$arrow.l(x)$` $\overline{xyz}$ `$overline(xyz)$` $\overbrace{xyz}$ `$overbrace(xyz)$` $\overbrace{xyz}$ `$overbracket(xyz)$`**7.2.2 Equals & Operators****Symbol Raw** $=$ `$=$` $\neq$ `$!=$` $\simeq$ `$tilde.eq$` $\geq$ `$gt.eq$` $\nless$ `$lt.eq.not$` $\not\approx$ `$approx.not$` $=:$ `$eq.colon$` $+$ `$plus$` $\oplus$ `$plus.circle$` $\mp$ `$minus.plus$`**Symbol Raw** $=$ `$eq$` $\equiv$ `$ident$` $\nless$ `$tilde.eq.not$` $\nless$ `$gt.eq.not$` $\approx$ `$approx$` $:$ `$colon$` $::=$ `$colon.double.eq$` $+$ `$plus.small$` $-$ `$-$` $\ominus$ `$minus.circle$`**Symbol Raw** $\neq$ `$eq.not$` $\equiv$ `$ident.strict$` $=$ `$eq.small$` $\leq$ `$lt.eq$` $\approx$ `$approx.eq$` $:=$ `$colon.eq$` $+$ `$+$` $\pm$ `$plus.minus$` $-$ `$minus$`**7.2.3 Scripts****Symbol Raw** x_1 `$x_1$` x_1 `$x_1$` x_1^2 `$x_1^2$` $\overset{2}{x}_1$ `$x_1^2$`**Symbol Raw** x_{12} `$x_(12)$` x_{12} `$x_(12)$` x_{12}^{34} `$x_(12)^(34)$` x_{12}^{34} `$x_(12)^(34)$`**Symbol Raw** x_1 `$scripts(x)_1$` x_1 `$scripts(x)_1$` x_1^2 `$scripts(x)_1^2$` x_1^2 `$scripts(x)_1^2$`**7.2.4 Special Elements****Symbol****Raw** $\binom{n}{k}$ (5) `$ binom(n, k) $` $\left[\frac{1}{2}\right]$ (7) `$ round(1, 2) $`**Symbol****Raw** $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (6) `$ vec(1, 2, delim: "[") $` $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (8) `$ mat(1,2; 3,4) $`

$$\begin{pmatrix} 1 & 2 & \dots & 10 \\ 2 & 2 & \dots & 10 \\ \vdots & \vdots & \ddots & \vdots \\ 10 & 10 & \dots & 10 \end{pmatrix} (9)$$

```
$ mat(
  1, 2, ..., 10;
  2, 2, ..., 10;
          dots.v,
dots.v, dots.down,
dots.v;
  10, 10, ...,
  10;
) $
```

$$\sum a_k \quad (10) \quad \$ \text{sum } a_k \$$$

$$\sum_{k=0}^n a_k \quad (11) \quad \$ \text{sum_}(k=0)^n a_k \$$$

$$\sum_{k=0}^n a_k \quad (12) \quad \$ \text{scripts}(\text{sum})_ (k=0)^n a_k \$$$

$$\sqrt[3]{x} \quad (13) \quad \$ \text{root}(3, x) \$$$

$$f(x, y) := \begin{cases} 1 & \text{if } \frac{x \cdot y}{2} \leq 0 \\ 2 & \text{if } x \text{ is even} \\ 3 & \text{if } x \in \mathbb{N} \\ 4 & \text{else} \end{cases} \quad (14)$$

```
$ f(x, y) := cases(
  1 "if" (x dot y)/2 <= 0,
  2 "if" x "is even",
  3 "if" x in NN,
  4 "else",
) $
```

$$\frac{1}{2} \quad (15) \quad \$ 1/2 \$$$

$$\frac{1}{2} \quad (16) \quad \$ \text{frac}(1,2) \$$$

$$\frac{x+1}{x+2} \quad (17) \quad \$ (x+1)/(x+2) \$$$

$$\frac{(x+1)}{(x+2)} \quad (18) \quad \$ ((x+1))/((x+2)) \$$$

$$\prod \quad (19) \quad \$ \text{product} \$$$

$$n! = \prod_{k=1}^n k \quad (20) \quad \$ \text{product_}(k=1)^n k \$$$

$$n! = \prod_{k=1}^n k \quad (21) \quad \$ n! = \text{scripts}(\text{product})_ (k=1)^n k \$$$

$$\int \quad (22) \quad \$ \text{integral} \$$$

$$\int_a^b f(x) \quad (23) \quad \$ \text{integral} \$$$

7.2.5 Alphabeth

Symbol

$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\omicron\rho\sigma\tau\upsilon\varphi\chi\psi\omega$

ΑΒΓΔΕΖΗΘΙΚΑΜΝΞΟΠΡΣΤΥΦΧΨΩ

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Raw

α β γ δ ϵ ζ η θ ι κ λ μ ν ξ $\omicron$ π ρ σ τ υ φ χ ψ ω

α β γ δ ϵ ζ η θ ι κ λ μ ν ξ $\omicron$ π ρ σ τ υ φ χ ψ ω

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

7.2.6 Logical

Symbol	Raw	Symbol	Raw	Symbol	Raw
$\wedge$	<code>\$and\$</code>	$\bigwedge$	<code>\$and.big\$</code>	$\&$	<code>\$amp\$</code>
$\vee$	<code>\$or\$</code>	$ $	<code>\$bar.v\$</code>	$*$	<code>\$ast.op\$</code>
$*$	<code>\$ast.basic\$</code>	$*$	<code>\$ast.low\$</code>	$\oplus$	<code>\$plus.circle\$</code>
$\oplus$	<code>\$plus.circle.big\$</code>				

7.2.7 Operators

Symbol	Raw	Symbol	Raw	Symbol	Raw
$\sin x$	<code>\$sin x\$</code>	$\cos x$	<code>\$cos x\$</code>	$\tan x$	<code>\$tan x\$</code>
$\arcsin x$	<code>\$arcsin x\$</code>	$\arccos x$	<code>\$arccos x\$</code>	$\arctan x$	<code>\$arctan x\$</code>
$\sinh x$	<code>\$sinh x\$</code>	$\cosh x$	<code>\$cosh x\$</code>	$\tanh x$	<code>\$tanh x\$</code>
$\arg x$	<code>\$arg x\$</code>	$\csc x$	<code>\$csc x\$</code>	$\deg x$	<code>\$deg x\$</code>
$\det x$	<code>\$det x\$</code>	$\dim x$	<code>\$dim x\$</code>	$\exp x$	<code>\$exp x\$</code>
$\bmod x$	<code>\$mod x\$</code>	$\inf x$	<code>\$inf x\$</code>	$\log x$	<code>\$log x\$</code>
$\lim x$	<code>\$lim x\$</code>	$\liminf x$	<code>\$liminf x\$</code>	$\limsup x$	<code>\$limsup x\$</code>
$\min x$	<code>\$min x\$</code>	$\max x$	<code>\$max x\$</code>	$\sup x$	<code>\$sup x\$</code>

7.2.8 Arrows

SymRaw	SymRaw	SymRaw
Arrows right		
$\rightarrow$ <code>\$arrow\$</code>	$\longrightarrow$ <code>\$arrow.long\$</code>	$\mapsto$ <code>\$arrow.bar\$</code>
$\mapsto$ <code>\$arrow.bar.long\$</code>	$\Rightarrow$ <code>\$arrow.double\$</code>	$\Rightarrow$ <code>\$arrow.double.long\$</code>
$\mapsto$ <code>\$arrow.double.bar\$</code>	$\mapsto$ <code>\$arrow.double.bar.long\$</code>	$\Rightarrow$ <code>\$arrow.quad\$</code>
$\Rightarrow$ <code>\$arrow.stroked\$</code>	$\Rightarrow$ <code>\$arrow.filled\$</code>	$\dashrightarrow$ <code>\$arrow.dashed\$</code>
$\curvearrowright$ <code>\$arrow.curve\$</code>	$\rightsquigarrow$ <code>\$arrow.squiggly\$</code>	$\looprightarrow$ <code>\$arrow.loop\$</code>
Arrows left		
$\leftarrow$ <code>\$arrow.l\$</code>	$\longleftarrow$ <code>\$arrow.l.long\$</code>	$\leftarrow$ <code>\$arrow.l.bar\$</code>
$\longleftarrow$ <code>\$arrow.l.bar.long\$</code>	$\leftarrow$ <code>\$arrow.l.double\$</code>	$\longleftarrow$ <code>\$arrow.l.double.long\$</code>
$\leftarrow$ <code>\$arrow.l.double.bar\$</code>	$\leftarrow$ <code>\$arrow.l.double.bar.long\$</code>	$\leftarrow$ <code>\$arrow.l.quad\$</code>
$\leftarrow$ <code>\$arrow.l.stroked\$</code>	$\leftarrow$ <code>\$arrow.l.filled\$</code>	$\dashleftarrow$ <code>\$arrow.l.dashed\$</code>
$\curvearrowleft$ <code>\$arrow.l.curve\$</code>	$\leftarrow$ <code>\$arrow.l.squiggly\$</code>	$\looparrowleft$ <code>\$arrow.l.loop\$</code>
Double Arrows Left Right		
$\leftrightarrow$ <code>\$arrow.l.r\$</code>	$\nleftrightarrow$ <code>\$arrow.l.r.not\$</code>	$\longleftrightarrow$ <code>\$arrow.l.r.long\$</code>
$\leftrightarrow$ <code>\$arrow.l.r.double\$</code>	$\longleftrightarrow$ <code>\$arrow.l.r.double.long\$</code>	$\nleftrightarrow$ <code>\$arrow.l.r.double.not\$</code>
$\leftrightarrow$ <code>\$arrow.l.r.stroked\$</code>	$\leftrightarrow$ <code>\$arrow.l.r.filled\$</code>	$\longleftrightarrow$ <code>\$arrow.l.r.wave\$</code>

Arrows Top `$arrow.t$` `$arrow.t.triple$` `$arrow.t.filled$` `$arrow.t.bar$` `$arrow.t.quad$` `$arrow.t.dashed$` `$arrow.t.double$` `$arrow.t.stroked$` `$arrow.t.curve$`**Arrows Bottom** `$arrow.b$` `$arrow.b.triple$` `$arrow.b.filled$` `$arrow.b.bar$` `$arrow.b.quad$` `$arrow.b.dashed$` `$arrow.b.double$` `$arrow.b.stroked$` `$arrow.b.curve$`**Double Arrows Top Bottom** `$arrow.t.b$` `$arrow.t.b.filled$` `$arrow.t.b.double$` `$arrow.t.b.stroked$`**Arrows Diagonal Top Right** `$arrow.tr$` `$arrow.tr.filled$` `$arrow.tr.double$` `$arrow.tr.hook$` `$arrow.tr.stroked$`**Arrows Diagonal Bottom Right** `$arrow.br$` `$arrow.br.filled$` `$arrow.br.double$` `$arrow.br.hook$` `$arrow.br.stroked$`**Arrows Diagonal Bottom Left** `$arrow.bl$` `$arrow.bl.filled$` `$arrow.bl.double$` `$arrow.bl.hook$` `$arrow.bl.stroked$`**Arrows Diagonal Top Left** `$arrow.tl$` `$arrow.tl.filled$` `$arrow.tl.double$` `$arrow.tl.hook$` `$arrow.tl.stroked$`**Double Arrows Diagonal** `$arrow.tl.br$` `$arrow.tr.bl$`**Other Arrows** `$arrow.cw$` `$arrow.ccw.half$` `$arrow.cw.half$` `$arrow.ccw$`**7.2.9 Angles****Symbol Raw** `$angle$` `$angle.acute$` `$angle.l$` `$angle.r.double$` `$angle.right.arc$` `$angle.spheric$`**Symbol Raw** `$angle.rev$` `$angle.arc$` `$angle.r$` `$angle.right$` `$angle.right.dot$` `$angle.spheric.rev$`**Symbol Raw** `$angle.acute$` `$angle.arc.rev$` `$angle.l.double$` `$angle.right.rev$` `$angle.right.sq$` `$angle.spheric.top$`

7.2.10 Cool Symbols

Symbol	Raw	Symbol	Raw	Symbol	Raw
@	<code>\$at\$</code>	%	<code>\$co\$</code>	©	<code>\$copyright\$</code>
©	<code>\$copyright.sound\$</code>	°C	<code>\$degree.c\$</code>	€	<code>\$euro\$</code>
\$	<code>\$dollar\$</code>	£	<code>\$pound\$</code>	₩	<code>\$won\$</code>
¥	<code>\$yen\$</code>	฿	<code>\$bitcoin\$</code>	°F	<code>\$degree.f\$</code>
!	<code>\$excl\$</code>	¡	<code>\$excl.inv\$</code>	!!	<code>\$excl.double\$</code>
!?	<code>\$excl.quest\$</code>	↯	<code>\$arrow.zigzag\$</code>	⊗	<code>\$ast.circle\$</code>
⚙	<code>\$ast.triple\$</code>	χ	<code>\$chi\$</code>	☒	<code>\$floral\$</code>
⚔	<code>\$maltese\$</code>	¶	<code>\$pilcrow\$</code>	ℏ	<code>\$planck\$</code>
♣	<code>\$suit.club\$</code>	◆	<code>\$suit.diamond\$</code>	♥	<code>\$suit.heart\$</code>
♠	<code>\$suit.spade\$</code>	△	<code>\$triangle.stroked.nested2\$</code>		

7.2.11 Style

Symbol	Raw	Symbol	Raw
$ABC123$	<code>\$sans(A B C 1 2 3)\$</code>	$\mathfrak{ABC123}$	<code>\$frak(A B C 1 2 3)\$</code>
$ABC123$	<code>\$mono(A B C 1 2 3)\$</code>	$\mathbb{ABC123}$	<code>\$bb(A B C 1 2 3)\$</code>
$\mathcal{ABC123}$	<code>\$cal(A B C 1 2 3)\$</code>		

Symbol	Raw
$\sum_{i \in \mathbb{N}} 1 + i$	<pre>#show math.equation: set text(font: "Fira Math") \$sum_(i in NN) 1 + i\$,</pre>

8 | Emoji Symbols

This is an incomplete list for all emoji goto [here](#)
If the emoji module is imported the #emoji can be removed

```
#import emoji: *
```

Sym	Raw	Sym	Raw
	#emoji.face		

```
#bibliography("../03-tail/bibliography.bib", style:"apa")  
#bibliography("../03-tail/bibliography.bib", style:"chicago-author-date")  
#bibliography("../03-tail/bibliography.bib", style:"chicago-notes")  
#bibliography("../03-tail/bibliography.bib", style:"ieee")  
#bibliography("../03-tail/bibliography.bib", style:"mla")
```

Bibliography

- [1] P. Fettke, “State-of-the-Art des State-of-the-Art,” *Wirtschaftsinformatik*, pp. 257–266, 2006, doi: 10.1007/s11576-006-0057-3.